

One rejection, end to end — the machine-readable audit trail

Gate 2 · zero-inflated variable (e.g. asthma hospitalisations) · Propose → Verify → Execute

1 PROPOSAL · simulated LLM (fed directly to Gate 2)

```
"proposed_method": "jenks",  
"proposed_breaks": [0.00, 1.53, 3.95, 68.88]  
  
# naïve Fisher-Jenks on the raw, zero-inflated values  
# Tier 1 is SIMULATED in V1 – the harness feeds this straight to Gate 2
```



profile + verify

2 GATE 2 VERDICT · verbatim from the emitted JSON trace

```
"gate": "G2",  
"diagnosis": "zero_inflated",  
"passed": false, # 49.8% of tracts are zero  
"prescribed_method": "manual_break_at_zero_then_fisher_jenks",  
"prescribed_breaks": [0.0, 7.86, 20.61, 68.88],  
"instruction": "Mandate explicit break at 0, then Fisher-Jenks  
on the non-zero tail.  
DO NOT propose alternative methods.  
Use these exact breaks."
```



MANDATE (prescribed breaks + code)

3 EXECUTED CODE · the mandate, transcribed

```
- breaks = [0.00, 1.53, 3.95, 68.88] # LLM proposal – REJECTED  
  
+ # MANDATED CLASSIFICATION – DO NOT MODIFY  
  
+ breaks = [0.0, 7.86, 20.61, 68.88] # prescribed by Gate 2  
  
+ classified = np.digitize(values, bins=breaks, right=True)
```

The LLM is reduced to a code-assembler: it may only splice in the prescribed constants.

Verbatim from output/traces/gate2_classification_trace.json (case: zero_inflated); regenerate with `autocarto demo`. Break values rounded for display — the trace stores full float precision. Re-running reproduces the trace byte-for-byte.